

This handout includes space for every question that requires a written response. Please feel free to use it to handwrite your solutions (legibly, please). If you choose to typeset your solutions, the `README.md` for this assignment includes instructions to regenerate this handout with your typeset L^AT_EX solutions.

1.a

The minimum cost of reaching (m, n) from $(0, 0)$ is

$$\frac{m(m+1)}{2} + n.$$

The cost only depends on the x being left, so take all n north steps while still at $x = 0$, where each costs 1 and they total n , then the m east steps leave from $x = 0, 1, \dots, m-1$ in turn and cost $1, 2, \dots, m$; writing that series forwards and backwards pairs it into m columns of $m+1$, so it sums to $m(m+1)/2$. One path achieving the minimum is to take all n steps north from the origin first and then all m steps east, and it is the only one, since any north step taken after moving east costs more than 1 because it leaves from $x > 0$, and a detour into negative x adds at least one step west and one step back east, each costing at least 1, without making any other step cheaper.

1.b

1. False. The states go on forever, but UCS takes them in cost order and every step costs at least 1, so anything cheaper than the goal is only so many steps out from the origin. There are only so many of those, so it gets to the goal and stops.
2. False. It does give back the minimum cost path, but it looks outside the box on the way. $(-1, 0)$ and $(0, -1)$ both cost 1 to get to, which is cheaper than most goals, so they get processed even though they sit outside $0 \leq x \leq m$, $0 \leq y \leq n$.
3. False. It returns the minimum cost path, but it can also look at places costing exactly the same as the minimum rather than less. When things tie, the queue decides who goes first. For the goal $(1, 0)$, which costs 1, the states $(-1, 0)$, $(0, -1)$ and $(0, 1)$ also cost 1 and come out before it.

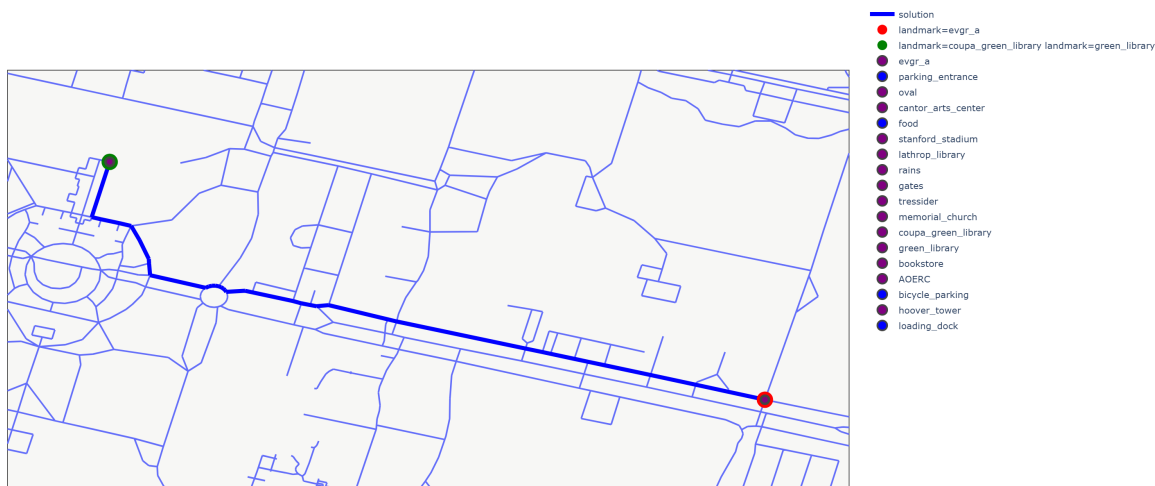
1.c

1. True. Adding a connection only gives you more paths, it never takes one away, so the minimum can drop or stay where it is but it cannot go up.
2. False. An action coming out of a state the start cannot get to is not on any path from start to goal, however cheap it is. Say $A \xrightarrow{5} C$ is the only way from start A to goal C , and $D \xrightarrow{-1000} E$ sits off on its own, then that cheap action never shows up. Same if it leads into a dead end.
3. False. Adding 1 to every action charges a path by how many edges it has, so the longer ones get hit harder and the order can flip. Route 1 is $A \xrightarrow{10} Z$ with one edge, route 2 is $A \xrightarrow{2} B \xrightarrow{2} C \xrightarrow{2} D \xrightarrow{2} Z$ with four. Before, route 1 costs 10 and route 2 costs 8, so route 2 wins. After, route 1 costs 11 and route 2 costs 12, so route 1 wins.

2.b

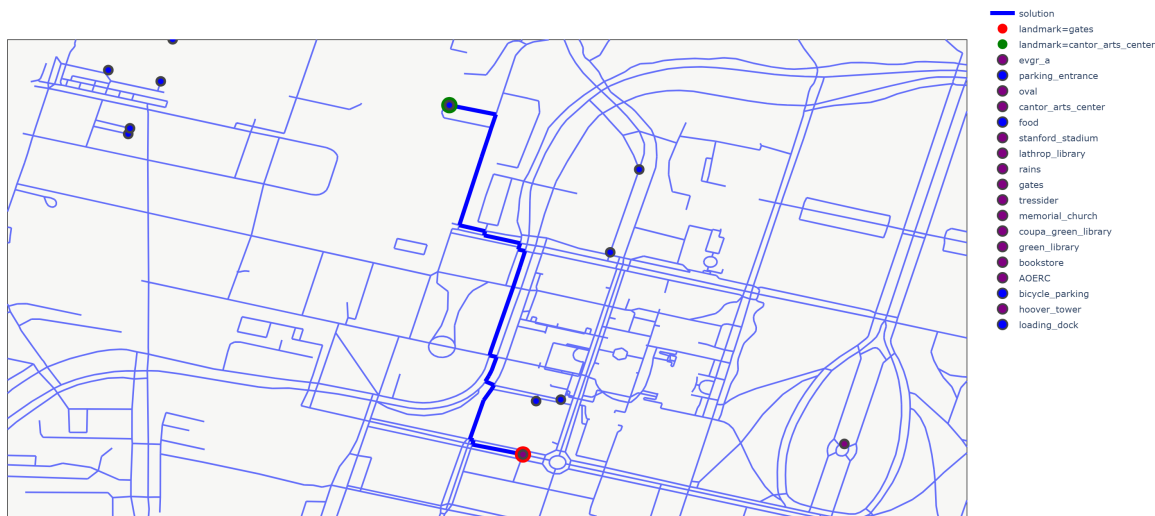
The first route start at EVGR-A with end tag `landmark=green_library`, which comes out around 606 m over 38 locations.

2b (useful): EVGR-A to Green Library



The second route start at Gates with end tag `amenity=food`, about 659 m, and it finishes at the Cantor Arts Center.

2b (undesirable): Gates to the nearest amenity=food



What I got out of it is that the cost is just distance, so anything that actually slows a walk down is free as far as the search is concerned. My EVGR-A route is around 606 m and goes through a couple of marked crossings, while the Gates one is roughly the same length but goes through about ten of them plus a bus stop, so the second walk takes a fair bit longer in practice and the search has no way to tell them apart. The locations are also spaced pretty unevenly, anywhere from under 2 m to nearly 100 m apart on the same path, so the number of steps does not say much about how far it is.

What seems wrong is that asking Gates for the nearest `amenity=food` sends to the Cantor Arts Center, while Tressider, which is where we would actually go, is further out and never gets picked. There are only a handful of locations on the map with `amenity=food` at all, and a couple of them are the art museum and the football stadium, which happens because the tag is sitting on the landmark's own node, so one cafe somewhere inside a big building ends up labelling the whole thing as food. The assumption that breaks is that a tag on one node stands for the whole place, and that somewhere you can walk to is somewhere that is open.

2.c

If everyone plans with the same system, the users all get sent down the same shortest path, and the congestion that creates never shows up in the cost, because the cost is only distance. The people who do not use it are hit whenever their own route intersects that optimal path, so quiet side streets that were never built for through traffic end up carrying it, and they get none of the benefit since they never opted in. The mismatch is that the model prices an edge as a fixed number when the real price depends on how many people are on it at the time, and the system has no way of knowing that its own advice is what put them there. One way to reduce that is to put actual speed into the cost rather than distance alone, so a road that is slow or already saturated costs more and the system spreads people out instead of funnelling them all onto one route.

3.b

The maximum number of states UCS could visit is

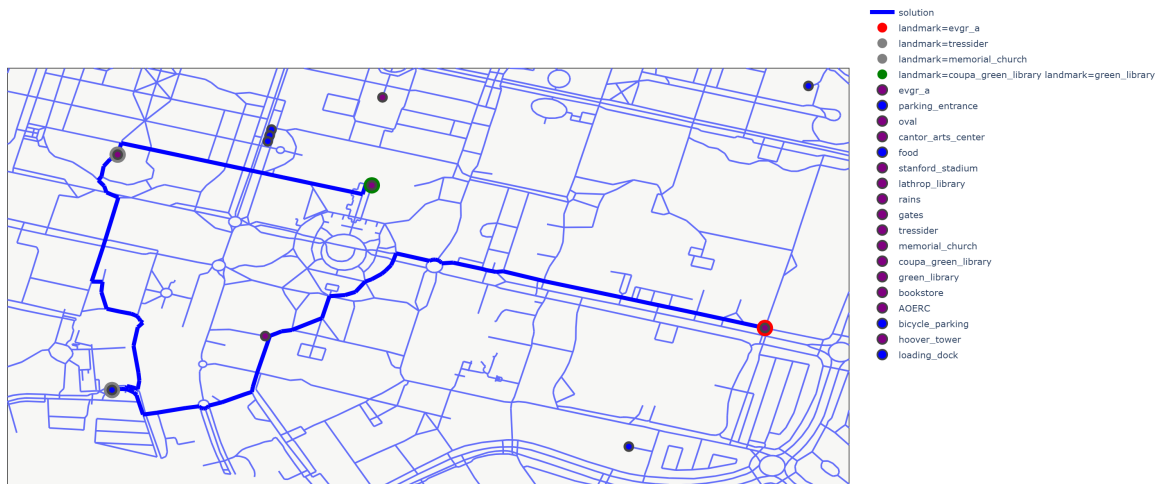
$$n \cdot 2^k.$$

A state is a location together with the set of waypoint tags covered so far, so there are n choices for the location, and for the memory each of the k tags is independently either covered or not, which gives 2^k possible sets. Multiplying the two independent choices gives $n \cdot 2^k$.

3.c

The waypoint tags are `landmark=memorial_church` and `landmark=tressider`, starting at `EVGR-A` and ending at `landmark=green_library`. It comes out around 1800 m over 130 locations.

3c: EVGR-A, via Memorial Church and Tressider, to Green Library



The order was not what I expected. I listed the church first but it went to Tressider first, then the church, then the library. That is the point of the waypoints being unordered, I do not have to pick the order myself, and my order would have cost more. The same trip with no waypoints is around 600 m, so two stops made it roughly three times longer, which makes sense because the stops sit off to the side.

What it misses is everything other than distance. For example, it does not know Tressider closes in the evening, or that I would want to eat before the library and not after.

3.d

Between a drop-off and the next pick-up I would insert a rest waypoint, tagged for a restroom, food, or somewhere to park for ten minutes, in the same way problem 4c joins a fake node to every end location at zero cost so the search finds whichever one is nearest. Here the cost would be small rather than zero, so a break is cheap enough that the route takes one but not so free that the driver is sent miles out of the way for it. To pick sensible waypoints I would want the driver's age and years of driving experience, since a new driver and a driver in their seventies do not need the same break pattern or the same length of shift. Those same two fields are exactly what a company could turn against them: knowing a driver's age lets dispatch quietly stop sending work to older drivers, which is age discrimination dressed up as routing, and it never has to be written down anywhere because it only shows up as fewer jobs.

5.a

Interactive tutoring session on Dynamic Programming:

<https://claude.ai/share/96fd17a6-37a0-4650-a6fc-3c82c68c815b>

5.b

Interactive tutoring session on Beam Search:

<https://claude.ai/share/c969fba9-4932-4670-9952-aa6afa912e3a>

6.a

My product is Claude Code, carried over from Assignment 1. It is marketed through non-consumer channels rather than advertising: new releases go out through Anthropic's own newsroom and social media and get picked up by news outlets from there, and the product page at claude.com/product/claude-code carries the pitch and the install line in the same place. Anthropic also runs its own developer conference, Code with Claude, which in 2026 ran in San Francisco, London and Tokyo. The other way in is work, since it is sold to businesses on Team and Enterprise plans (claude.com/pricing), so plenty of developers meet it because their employer signed an agreement rather than because they went looking. Past that it is mostly word of mouth in developer communities, and the brand level advertising, like the Super Bowl spots in February 2026, is for Claude as a whole and never mentions Claude Code.

6.b

Anthropic's mission on its company page is "Anthropic is an AI safety and research company. We build reliable, interpretable, and steerable AI systems" (anthropic.com/company), and I think the marketing lines up with it rather than overselling. The Life Sciences Verification Program gives verified researchers access to biology work the general models block (anthropic.com/news/life-sciences-verification-program), which is the steerable part of that mission shipped as a product. The September 2026 threat intelligence report goes the other way and documents real misuse of Claude across seven categories (anthropic.com/threat-intelligence-report-september-2026), which is not something you publish if you are exaggerating what your models can do. So I would call it fair rather than sensationalised.

6.c

Support is written only. The help page states plainly that "We don't offer phone support" and that "Support is provided in writing through the support messenger and email", with an AI bot called Fin answering first and a human product support team reachable only on Pro, Max, Team and Enterprise, while free users get the documentation and the bot (support.claude.com). The process is asynchronous by design, so a specialist "investigates and replies in the same conversation" with response times varying by plan and severity, and there is no realtime phone or video route at all.

6.d

Both queries went through the support messenger at support.claude.com.

Query 1, sent about Claude Code:

I use Claude Code on a Max plan and I'd like to know whether it can be pointed at a different model endpoint. I know Bedrock and Vertex are supported. Is there any supported way to use an OpenAI-compatible or OpenRouter endpoint, and if not, is that something on the roadmap? I'm asking because I want to compare outputs on the same repo without changing tools.

The reply:

Claude Code can connect through LLM gateways that expose a supported API format, but we don't support routing Claude Code to non-Claude models through any gateway.

While you can configure Claude Code to use different Claude model endpoints (like Bedrock, Vertex, Microsoft Foundry, or through custom `ANTHROPIC_BASE_URL` settings), the gateway compatibility is specifically for accessing Claude models, not for switching to other model providers like OpenAI or OpenRouter.

For comparing outputs on the same repo, you'd need to use separate tools for non-Claude models. We don't have information about future roadmap plans for multi-model support in Claude Code.

Query 2, sent about Claude itself:

I'm using Claude to learn the maths for a machine learning course, and I want it to tutor me rather than solve things for me. The problem I keep hitting is that when I'm deriving something, Claude already knows the final form and reasons backwards toward it, which looks like a derivation but doesn't help, because the answer is the one thing I don't have. Telling it to stop works for a few messages and then it drifts back. Where is the best place to put an instruction like that so it holds for a whole session, Project instructions, a custom style, or something else? And is there a recommended way to phrase it?

The reply:

For persistent tutoring instructions, Project instructions are your best option. They apply to all chats within that project, so you can keep your ML course work in one project with the tutoring behavior locked in.

Here's how to phrase it effectively. In your project instructions, try something like: "Act as a tutor, not a solver. When I'm deriving equations or working through problems, guide me with hints and questions rather than showing the final form or working backwards from the answer. Let me discover the solution step-by-step. If I get stuck, ask what I've tried or suggest the next small step, but don't reveal the destination."

You could also add this to your account-wide Instructions for Claude (Settings → Instructions for Claude) if you want this tutoring approach across all your conversations, not just in one project. Project instructions work on top of your account instructions, so you can combine both, general preferences account-wide and specific tutoring rules in your ML project.

Both queries were returned through the online fin agent without human replies.